Lab06 Exercises

❖ Lab06.1. Overriding methods of a Math class

Write a program to create a class “Math” and inherit from following “AbstractMath”
After all, override all functions and implement its own body statement to calculate accordingly.

*** You are not allowed to use existing libraries or classes in Java

```java
class AbstractMath {
    int reverseNumber(int num) {
        // override to return a reverse number
        // ex. 12345 to 54321
        return 0;
    }

    int roundedNumber(double num) {
        // override to return a arounded number
        // ex1. 3.14 to 3
        // ex2. 3.5 to 4
        // ex3. 3.7 to 4
        return 0;
    }

    int countDistinct(int[] values) {
        // override to return the count of unique value
        // ex1. { -1, -1, 1, 0, 1, 5, 9 } -> 3
        // ex2. { -5, 1, 0, 6, 5, 0 } -> 4
        return 0;
    }

    int decimalToBinary(int decimal) {
        // override to return the binary
        // coverted from decimal
        // ex1. 27 -> 11011
        // ex2. 121 -> 1111001
        return 0;
    }

}
```

Console:

```
==== Menu ===
1.  Reverse a number
2.  Rounded number
3.  Count distinct number
4.  Decimal to Binary
5.  Quit
Choose an opt: 1

::: Reverse a number :::
Input number: 4567
Output is: 7654
```